ECE 3210 hw07

1. These are old exam questions. They are easy to check numerically, so you should verify your work with Python or NumPy. However, you must still be able to do these calculations by hand on an exam.

   (a) Given the discrete sequences

   $$f = \{1, 1, 1, 1, 0, 0, 0\}, \qquad g = \{1, 2, 3, 4, 3, 2, 1\},$$

   find the 7-point circular convolution of f and g. Represent your answer as a vector or array.

   (b) Using the DFT, verify your answer by computing the FFT of both sequences, multiplying them, and reporting the inverse FFT result. Do this using the real part only if the result is complex-valued.

   (c) What length of circular convolution is required to compute the linear convolution of f and g? Express your answer as a single integer.

2. Consider the following sequences:

$$f_k = \{-1, 0, 1, -1, 0, 1\}, \qquad g_k = \{1, 1, 1, 1, 1, 1\}.$$

(a) Find the linear convolution $y_{k,\text{lin}}$ of f_k and g_k.

(b) Now find the circular convolution of f_k and g_k.

3. Consider the discrete-time signal

$$x[n] = \begin{cases} 1 + \cos\left(\frac{\pi n}{4}\right) - 0.5\cos\left(\frac{3\pi n}{4}\right), & 0 \leq n \leq 7, \\ 0, & \text{otherwise.} \end{cases}$$

Compute the 8-point DFT $X_8[k]$ of $x[n]$, and plot $X_8[k]$ for $0 \leq k \leq 7$.

4. (a) Suppose you have a signal $x(t)$ that you sample at 200 Hz to obtain a 20-point sequence x_k. If you take a 20-point DFT of this sequence, what frequencies are represented by the DFT output? Give your answer in Hz.

   (b) Now suppose you zero-pad this sequence x_k to have 30 samples and take a 30-point DFT. What frequencies are represented in that DFT? Give your answer in Hz.

Answer Key

1(a) $y = [7, 6, 7, 10, 12, 12, 10]$
1(b) $y_{\text{FFT}} = [7, 6, 7, 10, 12, 12, 10]$
1(c) $N = 13$
2(a) $y_{\text{lin}} = [1, 3, 6, 10, 12, 12, 10, 6, 3, 1]$
2(b) $y_{\text{circ}} = [0, 0, 0, 0, 0, 0]$
3(b) $X_8 = [8, 4, 0, -2, 0, -2, 0, 4]$
4(a) $f = [0, 10, 20, 30, \ldots, 190]$ Hz
4(b) $f = [0, 6.67, 13.33, 20, \ldots, 193.33]$ Hz